

		Management	
2	TLO 2.1 Explain the different states of a process. TLO 2.2 Describe the functions of different component of process stack in PCB (Process Control Block). TLO 2.3 Explain multiple processes access shared resources without interfering each other. TLO 2.4 Compare Multithreading models.	Unit - II Process Management 2.1 Processes: process state, process control block 2.2 Process Scheduling: scheduling queues, types of schedulers, context switch 2.3 Inter Process Communication: Shared memory system, Message passing system 2.4 Threads: Benefits, User and Kernel level threads, Multithreading Models: One to One, Many to One, Many to Many 2.5 Execute process commands like: top, ps, kill, wait, sleep, exit, nice	14 marks Lecture Using Chalk-Board Presentations

Introduction - Process management.

- ↳ The operating system manages many kinds of activities ranging from user operating programs to system programs.
- ↳ The major activities/functions of an operating system in regard to process management are:
 - 1) Creation and deletion of user and system processes.
 - 2) Suspension and resumption of processes.
 - 3) A mechanism for process synchronization, process communication and deadlock handling.

Program & Process

Program: A program is series or set of instructions to perform a particular task.

Process: A process is a program in execution which competes for CPU time & other resources.

<u>Program</u>	<u>Process</u>
It is passive entity	It is active entity
Program is static	Process is dynamic
It does not have any control block	Process has own control block called PCB.
It contains instructions.	It is sequence of instruction execution
Time span is unlimited	Time span is limited
Program is loaded in Secondary storage	It is loaded in main memory

Process state

Process is program in execution.

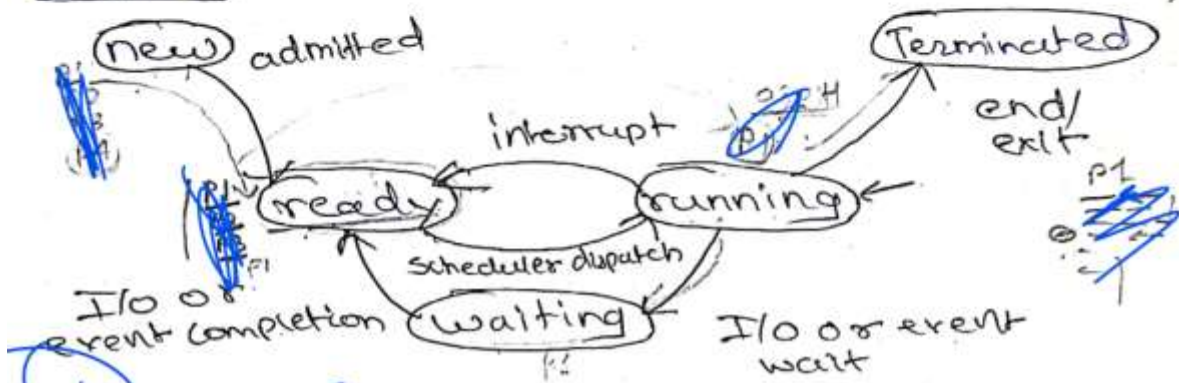
Process from creation to end or execute the process goes through five states that is Process State.

As process executes, it changes state.

Different process state are as follows.

- 1) New
- 2) Ready
- 3) Running
- 4) Waiting
- 5) Terminated

Diagram



New state: When a ~~process~~ process enters into the system it is in new state.

In this process state, process is created. In new state process in job pool.

Ready state: When the process is loaded into the main memory, it is ready for execution.

In this state the process is waiting for processor allocation.

Running: When CPU is available, system selects one process from main memory and executes all the instructions from that process. So, when process is in execution, it is in running state.

Single-user system: 1 process in running state.
Multi-user system: Multiple process in running state.

Waiting State : When process is in execution, it may request for I/O resources. If resources are not available, process goes in waiting state. When resources are available, then process goes back to ready state.

Terminated State : When the process completes its execution, it goes into the terminated state. In this state memory occupied by process is released.

Process Control Block

process is released.

Process Control Block (PCB)

Process is defined as program in execution, which competes for CPU time and other resources. Process is also called as job, task or unit of work.

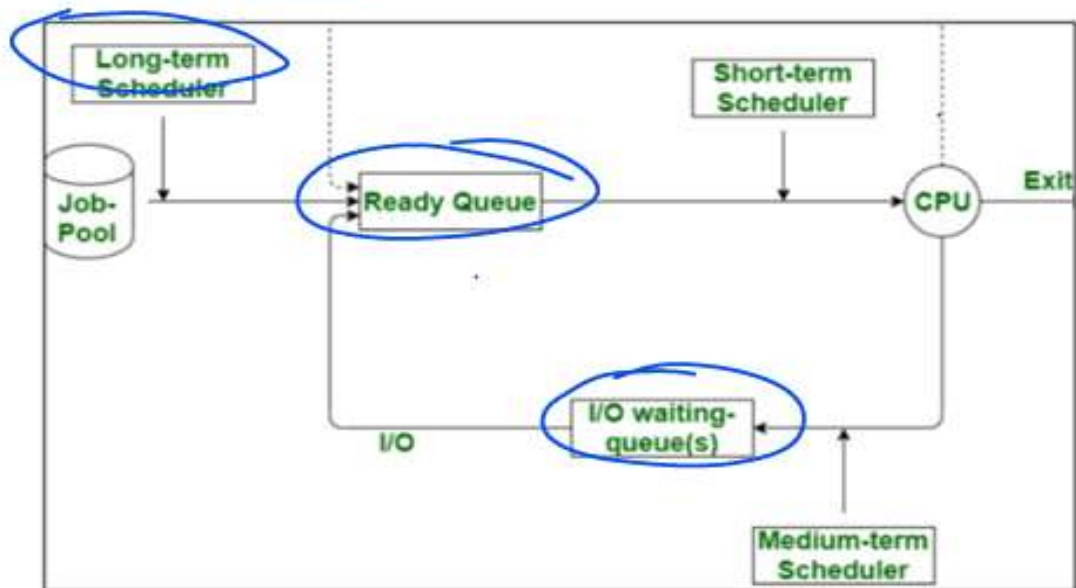
Process Control Block

Process control block is data structure that contains information of process related to it. The process control block is also known as Task control block.

Pointer	Process State
Process Number (PID)	
Program Counter	
CPU Registers or Registers	
Memory limits	
List of open files	
Board of Tech	
Fig. 1.1 PCB	

- 1) Pointer: Pointer points to another process control block. Pointer is used for maintaining scheduler list.
- 2) Process state: This information is about the current state of the process. The state may be new, ready, running, and waiting, halted and so on.
- 3) Process Number: Each process is identified by its process number called Process Identification Number (PID). Every process has unique process-id through which it is identified. It is provided by OS. Process id of two process cannot be same as its unique.
- 4) Program Counter: The counter indicates the address of next instruction to be executed for this process.
- 5) CPU Registers: Various CPU registers are there where process need to be stored for execution for running state. include, accumulators, index registers, stack pointers and General purpose registers.
- 6) Process Privileges: This is required to allow/denow access to system resources.
- 7) CPU scheduling Information: Process priority and other scheduling information which is required to schedule the process.
- 8) Memory management Information: This includes the information of page table, memory limits, Segment table depending on memory used by OS.
- 9) Accounting Information: This includes the amount of CPU used for process execution, time limits etc ---
- 10) I/O status Info: This includes a list of I/O device allocated to process.

SCHEDULER / QUEUES



Long-term scheduler:

- It picks programs from a big pool (job pool) and puts them into the main memory.
- It decides how many programs can run at once (this is called the degree of multiprogramming).
- A program can be:
 - I/O-bound (spends more time doing input/output tasks).
 - CPU-bound (spends more time doing calculations).
- The scheduler tries to balance both types to keep the system running smoothly.
- It runs rarely, only when a program finishes and leaves the system.
- When it picks a program, the program's state changes from new to ready.

Short-term scheduler:

- Also called the CPU scheduler.
- It picks a program from the ready queue (programs waiting to run) and gives it to the CPU.
- This scheduler works often because the CPU needs tasks quickly.
- When it picks a program, the program's state changes from ready to running.

Medium-term scheduler:

- If a program is running but gets blocked (e.g., waiting for input), it is swapped out (moved to storage).
- When there's space in the main memory, this scheduler picks a program from storage and puts it back in the ready queue.
- It works closely with the long-term scheduler to decide what to load into the memory.

QUEUES THEORY GIVEN-----